Related Work Sources for a UAV Orthomosaic → Temporally Stable Tree Crowns → Crown-Level Phenology Pipeline

TL;DR

- This is a ranked, six-theme shortlist of **34 real, verifiable sources**, each with authors, year, venue, a one-sentence summary, a relation line to your pipeline, and a complete BibTeX entry; every named seminal work you requested is included and confirmed (Detectree2/Ball 2023, DeepForest/Weinstein 2020, Mask R-CNN/He 2017, Guizar-Sicairos 2008, SORT/Bewley 2016, DeepSORT/Wojke 2017, Kuhn 1955, PhenoCam/Richardson 2018, Sonnentag 2012, Woebbecke 1995, SeasonWatch/Ramaswami 2021, USA-NPN/Schwartz 2012).
- The strongest framing of your novelty: existing UAV crown-phenology studies (Klosterman & Richardson 2017, Wu et al. 2021) draw ROIs once and treat them as static across dates, and tracking-by-detection (SORT/DeepSORT) leans on the Hungarian one-to-one assignment (Kuhn 1955) that breaks under split/merge — **your graph-based association producing medoid "consensus crowns" is the gap-filling contribution**, and Theme 2.4 should be written to make that contrast explicit.
- A small number of fields are flagged **[verify]** (mainly exact page ranges and a few contemporary-paper author lists) — confirm these against the publisher DOI page before camera-ready; do not ship them unverified.

THEME 2.1 — UAV-based vegetation and tree monitoring

1. Klosterman & Richardson (2017) — *Sensors* (MDPI). Summary: Tracks spring and fall phenology in a deciduous forest from repeat aerial drone RGB imagery, drawing crown- and grid-level ROIs and fitting sigmoid curves to GCC/redness time series to extract transition dates. Relation (same approach, your work fills the gap): This is the closest precedent to your pipeline — same drone-RGB, same GCC-on-crown-ROI logic — but ROIs are **manually drawn on a single date and assumed static**; you automate crown delineation (Detectree2), co-register dates, and temporally link crowns into consensus ROIs, directly addressing its temporal-stability limitation.

bibtex

```
@article{klosterman2017observing,
  author  = {Klosterman, Stephen and Richardson, Andrew D.},
  title   = {Observing Spring and Fall Phenology in a Deciduous Forest with Aeri
  journal = {Sensors},
  year    = {2017},
  volume  = {17},
  number  = {12},
  pages   = {2852},
  doi     = {10.3390/s17122852}
}
```

2. Berra, Gaulton & Barr (2019) — *Remote Sensing of Environment*. Summary: Compares ground, UAV, and Landsat observations of temperate-woodland spring phenology and shows UAV-derived greenness (GCC from the brightest pixels) minimizes branch-gap/interstitial confounds at fine scale. Relation (same approach, multiscale): Validates UAV RGB greenness against both ground and satellite — supports your UAV-vs-field-vs-satellite motivation; differs in that it uses fixed plots rather than automatically linked individual crowns. **[verify exact volume/pages — likely vol. 223]**

bibtex

```
@article{berra2019assessing,
  author  = {Berra, Elias F. and Gaulton, Rachel and Barr, Stuart},
  title   = {Assessing spring phenology of a temperate woodland: A multiscale co
  journal = {Remote Sensing of Environment},
  year    = {2019},
  volume  = {223},
  pages   = {229--242},
  doi     = {10.1016/j.rse.2019.01.010},
  note    = {Verify volume/pages/DOI against publisher}
}
```

3. Ecke et al. (2022) — *Remote Sensing* (MDPI). Summary: Systematic review of UAV-based forest-health monitoring covering RGB/multispectral/hyperspectral/LiDAR sensors and workflows, noting that single-date drone data capture only a momentary condition. Relation (review backbone): Provides the UAV-vs-traditional-methods trade-off framing and explicitly motivates **repeat/multi-temporal flights** — the premise of your ~2-week cadence across the year.

bibtex

```
@article{ecke2022uav,
  author  = {Ecke, Simon and Dempewolf, Jan and Frey, Julian and Schwaller, Andr
  title   = {UAV-Based Forest Health Monitoring: A Systematic Review},
  journal = {Remote Sensing},
  year    = {2022},
  volume  = {14},
  number  = {13},
  pages   = {3205},
  doi     = {10.3390/rs14133205}
}
```

4. Urban tree UAV multi-sensor classification (2021) — *GIScience & Remote Sensing*. Summary: Fuses UAV multispectral, hyperspectral, LiDAR, and thermal imagery to classify urban tree species with limited training samples, emphasizing UAV cost/resolution advantages and small-area coverage limits relative to satellite/airborne data. Relation (urban setting): Directly supports your **urban context** (IIT Delhi campus) and the UAV-vs-satellite trade-off discussion. **[verify author list against tandfonline DOI 10.1080/15481603.2021.1974275]**

bibtex

```
@article{urban2021uavmultisensor,
  author  = {{[verify authors]}},
  title   = {Urban tree species classification using UAV-based multi-sensor data
  journal = {GIScience \& Remote Sensing},
  year    = {2021},
  volume  = {58},
  number  = {8},
  pages   = {1250--1275},
  doi     = {10.1080/15481603.2021.1974275},
  note    = {Verify authors and pages against publisher}
}
```

5. UAV remote-sensing platforms/sensors review (2023) — *Drones* (MDPI). Summary: Broad review of UAV remote-sensing platforms, sensor categories, and data-processing pipelines (including orthomosaic generation) and their applications. Relation (methodological grounding): General reference for your **WebODM orthomosaic** workflow and sensor choices. **[verify author list against MDPI Drones 7(6):398]**

bibtex

```
@article{drones2023uavreview,
  author  = {{[verify authors]}},
  title   = {A Review on Unmanned Aerial Vehicle Remote Sensing: Platforms, Sens
  journal = {Drones},
  year    = {2023},
  volume  = {7},
  number  = {6},
  pages   = {398},
  doi     = {10.3390/drones7060398},
  note    = {Verify authors against publisher}
}
```

THEME 2.2 — Individual tree crown detection and delineation (ITCD)

1. Ball et al. (2023) — *Remote Sensing in Ecology and Conservation* (Detectree2). **[Seminal — your core detector]** Summary: Introduces Detectree2, a Mask R-CNN/Detectron2 instance-segmentation model that delineates individual tree-crown **polygons** from airborne RGB; per the paper, "Detectree2 delineated 65 000 upper-canopy trees across 14 km², F1 = 0.64 ... and for the tallest category of trees ... F1 = 0.74." Relation (same approach): This is the crown detector in your pipeline — you extend it with **multi-confidence-threshold tiled inference** and downstream temporal linking. Plymsea

bibtex

```
@article{ball2023accurate,
  author  = {Ball, James G. C. and Hickman, Sebastian H. M. and Jackson, Tobias
  title   = {Accurate delineation of individual tree crowns in tropical forests
  journal = {Remote Sensing in Ecology and Conservation},
  year    = {2023},
  volume  = {9},
  number  = {5},
  pages   = {641--655},
  doi     = {10.1002/rse2.332}
}
```

2. Weinstein et al. (2020) — *Methods in Ecology and Evolution* (DeepForest). **[Seminal]** Summary: Open-source Python package using a RetinaNet detector pretrained on LiDAR-derived labels and fine-tuned on RGB to predict tree **bounding boxes**, with a `predict_tile` windowed-inference/NMS routine for large images. Relation (differs — boxes vs masks): Foundational ITCD baseline and the origin of your **tiled-inference** strategy; your work deliberately requires **polygon masks** (Detectree2) rather than DeepForest boxes, and you make the explicit argument that masks are necessary for clean crown-level phenology crops.

bibtex

```
@article{weinstein2020deepforest,
  author  = {Weinstein, Ben G. and Marconi, Sergio and Aubry-Kientz, M{\'e}laine
  title   = {DeepForest: A Python package for RGB deep learning tree crown delir
  journal = {Methods in Ecology and Evolution},
  year    = {2020},
  volume  = {11},
  number  = {12},
  pages   = {1743--1751},
  doi     = {10.1111/2041-210X.13472}
}
```

3. He et al. (2017) — *ICCV* (Mask R-CNN). **[Seminal architecture]** Summary: General instance-segmentation framework extending Faster R-CNN with a parallel mask branch and RoIAlign, winner of the 2017 ICCV Marr Prize. Relation (foundation): The architecture beneath Detectree2 — cite as the methodological basis for why pixel-accurate instance **masks** (not just boxes) are obtainable for each crown. TheCVF IEEE Signal Processing Soci...

bibtex

```
@inproceedings{he2017mask,
  author    = {He, Kaiming and Gkioxari, Georgia and Doll{\'a}r, Piotr and Girsh
  title     = {Mask R-CNN},
  booktitle = {Proceedings of the IEEE International Conference on Computer Visi
  year      = {2017},
  pages     = {2961--2969},
  doi       = {10.1109/ICCV.2017.322}
}
```

4. Weinstein et al. (2019) — *Remote Sensing* (semi-supervised crown detection). Summary: Shows semi-supervised pretraining on LiDAR-generated noisy labels followed by RGB fine-tuning improves crown-detection generalization across NEON sites. Relation (supports design): Justifies your **confidence-thresholding and tiling** choices for generalizing a deep detector across sites; box-based, so it differs from your mask requirement.

bibtex

```
@article{weinstein2019individual,
  author  = {Weinstein, Ben G. and Marconi, Sergio and Bohlman, Stephanie and Za
  title   = {Individual Tree-Crown Detection in RGB Imagery Using Semi-Supervise
  journal = {Remote Sensing},
  year    = {2019},
  volume  = {11},
  number  = {11},
  pages   = {1309},
  doi     = {10.3390/rs11111309}
}
```

5. Review of ITCD from optical remote sensing (2023) — preprint/review. Summary: Surveys classical (watershed, template-matching) and deep-learning ITCD methods across sensors and forest types. Relation (positions your choice): Frames your selection of a **mask-based deep ITCD** model within the broader detection/delineation literature. **[verify final venue — currently arXiv:2310.13481]**

bibtex

```
@article{itcd2023review,
  author  = {{[verify authors]}},
  title   = {A review of individual tree crown detection and delineation from op
  journal = {arXiv preprint arXiv:2310.13481},
  year    = {2023},
  note    = {Verify final published venue and authors}
}
```

THEME 2.3 — Multi-temporal orthomosaic alignment / co-registration

1. Guizar-Sicairos, Thurman & Fienup (2008) — *Optics Letters* (subpixel registration). **[Seminal — your registration core]** Summary: Efficient matrix-multiply DFT upsampling for **subpixel image registration by cross-correlation**, achieving FFT-upsampling accuracy at a small fraction of the computation/memory. Relation (same approach): This is the exact phase-cross-correlation method behind your **tiled/windowed co-registration** step; you apply the recovered shift to crown polygons.

bibtex

```
@article{guizar2008efficient,
  author  = {Guizar-Sicairos, Manuel and Thurman, Samuel T. and Fienup, James R.
  title   = {Efficient subpixel image registration algorithms},
  journal = {Optics Letters},
  year    = {2008},
  volume  = {33},
  number  = {2},
  pages   = {156--158},
  doi     = {10.1364/OL.33.000156}
}
```

2. Zitová & Flusser (2003) — *Image and Vision Computing* (registration survey). **[Foundational]** Summary: Canonical survey classifying image registration into area-based vs feature-based methods across four steps: feature detection, matching, transform-model design, and resampling. Relation (foundation): The framing reference for situating your registration design (area-based phase correlation) against feature-based alternatives. ResearchGate

bibtex

```
@article{zitova2003image,
  author  = {Zitov{\'a}, Barbara and Flusser, Jan},
  title   = {Image registration methods: a survey},
  journal = {Image and Vision Computing},
  year    = {2003},
  volume  = {21},
  number  = {11},
  pages   = {977--1000},
  doi     = {10.1016/S0262-8856(03)00137-9}
}
```

3. Nota, Nijland & de Haas (2022) — *International Journal of Applied Earth Observation and Geoinformation* (UAV-SfM co-alignment). Summary: Assesses 24 UAV-SfM workflows and shows that, in their words, "co-alignment reduces relative errors to below 2 cm regardless of positioning quality," recommending co-alignment for all change-detection studies. Relation (motivates GCP-free step): Direct empirical justification for your **GCP-free co-registration** of repeat orthomosaics; you operate on the finished orthomosaics with phase correlation rather than re-aligning the SfM block, a lighter-weight variant. ScienceDirect

bibtex

```bibtex
@article{nota2022improving,
  author  = {Nota, Ed{\'e}r W. and Nijland, Wiebe and de Haas, Tjalling},
  title   = {Improving UAV-SfM time-series accuracy by co-alignment and contribu
  journal = {International Journal of Applied Earth Observation and Geoinformati
  year    = {2022},
  volume  = {109},
  pages   = {102772},
  doi     = {10.1016/j.jag.2022.102772}
}
```

4. Feurer & Vinatier (2018) — *ISPRS Journal of Photogrammetry and Remote Sensing* (single-block multi-epoch SfM). Summary: Demonstrates that processing multiple-epoch images as a single SfM block enables GCP-free 3-D change detection using almost exclusively image information. Relation (conceptual basis): Underpins the idea that **residual coordinate drift** between consumer-drone epochs can be corrected without ground control — the problem your step targets. **[verify exact pages]**

bibtex

```bibtex
@article{feurer2018joining,
  author  = {Feurer, Denis and Vinatier, Fabrice},
  title   = {Joining multi-epoch archival aerial images in a single SfM block al
  journal = {ISPRS Journal of Photogrammetry and Remote Sensing},
  year    = {2018},
  volume  = {146},
  pages   = {495--506},
  doi     = {10.1016/j.isprsjprs.2018.10.016},
  note    = {Verify pages/DOI against publisher}
}
```

5. Rublee, Rabaud, Konolige & Bradski (2011) — *ICCV* (ORB). **[Seminal feature method]** Summary: Per the paper, ORB is "a very fast binary descriptor based on BRIEF ... rotation invariant and resistant to noise ... two orders of magnitude faster than SIFT, while performing as well in many situations." Relation (alternative you contrast): Represents the **feature-based (ORB) registration alternative**; you can position it (with ECC) as a baseline/fallback that differs from your chosen phase-correlation approach, which is more robust under low-texture canopy scenes. ACM Digital Library SciSpace

bibtex

```
@inproceedings{rublee2011orb,
  author    = {Rublee, Ethan and Rabaud, Vincent and Konolige, Kurt and Bradski,
  title     = {ORB: An efficient alternative to SIFT or SURF},
  booktitle = {Proceedings of the IEEE International Conference on Computer Visi
  year      = {2011},
  pages     = {2564--2571},
  doi       = {10.1109/ICCV.2011.6126544}
}
```

6. James, Robson & Smith (2017) — *Earth Surface Processes and Landforms* (SfM change-detection uncertainty). Summary: Quantifies how georeferencing strategy (GCP vs direct) drives precision and residual spatial offsets in SfM-based topographic change detection. Relation (evidence for the problem): Documents the **residual offset / coordinate-drift** phenomenon your co-registration corrects before change/phenology analysis.

bibtex

```
@article{james2017three,
  author  = {James, Mike R. and Robson, Stuart and Smith, Mark W.},
  title   = {3-D uncertainty-based topographic change detection with structure-f
  journal = {Earth Surface Processes and Landforms},
  year    = {2017},
  volume  = {42},
  number  = {12},
  pages   = {1769--1788},
  doi     = {10.1002/esp.4125}
}
```

THEME 2.4 — Object association and tracking-by-detection

1. Bewley, Ge, Ott, Ramos & Upcroft (2016) — *IEEE ICIP* (SORT). **[Seminal — the baseline you depart from]** Summary: Tracking-by-detection combining a constant-velocity Kalman filter with **Hungarian-algorithm IoU assignment**; extremely fast (~260 Hz) but, by design, ignores corner cases and suffers identity switches and fragmentation under occlusion. Relation (differs — you avoid strict Hungarian): The canonical reference establishing the frame-to-frame Hungarian assignment paradigm your **graph-based association explicitly rejects** in favor of handling one-to-one, containment, and split-merge. Roboflow + 3

bibtex

```
@inproceedings{bewley2016simple,
  author    = {Bewley, Alex and Ge, Zongyuan and Ott, Lionel and Ramos, Fabio an
  title     = {Simple online and realtime tracking},
  booktitle = {2016 IEEE International Conference on Image Processing (ICIP)},
  year      = {2016},
  pages     = {3464--3468},
  doi       = {10.1109/ICIP.2016.7533003}
}
```

2. Wojke, Bewley & Paulus (2017) — *IEEE ICIP* (DeepSORT). **[Seminal]** Summary: Extends SORT with a pretrained deep appearance (re-identification) metric to reduce identity switches through longer occlusions, while retaining the Kalman+Hungarian core. Relation (differs): Same tracking-by-detection lineage; you replace its appearance-metric + Hungarian matching with **IoU/overlap/centroid-distance/shape graph features** suited to slowly drifting, occasionally split/merged tree crowns observed weeks apart rather than video frames. Hugging Face

bibtex

```
@inproceedings{wojke2017simple,
  author    = {Wojke, Nicolai and Bewley, Alex and Paulus, Dietrich},
  title     = {Simple online and realtime tracking with a deep association metri
  booktitle = {2017 IEEE International Conference on Image Processing (ICIP)},
  year      = {2017},
  pages     = {3645--3649},
  doi       = {10.1109/ICIP.2017.8296962}
}
```

3. Kuhn (1955) — *Naval Research Logistics Quarterly* (Hungarian method). **[Seminal assignment foundation]** Summary: The foundational polynomial-time algorithm for the **one-to-one assignment problem** between two equal-size sets. Relation (the exact limitation you target): Its strict bijective assignment is precisely what fails on crown **split/merge, containment, and missed detections** across dates — the failure mode your work is built to overcome. Wiley Online Library

bibtex

```
@article{kuhn1955hungarian,
  author  = {Kuhn, Harold W.},
  title   = {The Hungarian method for the assignment problem},
  journal = {Naval Research Logistics Quarterly},
  year    = {1955},
  volume  = {2},
  number  = {1-2},
  pages   = {83--97},
  doi     = {10.1002/nav.3800020109}
}
```

4. Munkres (1957) — *Journal of the SIAM* (Kuhn–Munkres). Summary: Refines and proves correctness of the assignment/transportation algorithm now called Kuhn–Munkres. Relation (completes the assignment pair): Cite alongside Kuhn as the algorithmic basis you contrast against; reinforces that optimal **bijective** assignment is the wrong model for your many-to-one/one-to-many crown linking. arxiv

bibtex

```
@article{munkres1957algorithms,
  author  = {Munkres, James},
  title   = {Algorithms for the assignment and transportation problems},
  journal = {Journal of the Society for Industrial and Applied Mathematics},
  year    = {1957},
  volume  = {5},
  number  = {1},
  pages   = {32--38},
  doi     = {10.1137/0105003}
}
```

5. Zhang, Li & Nevatia (2008) — *IEEE CVPR* (global data association via network flows). **[Closest methodological analog]** Summary: Formulates multi-object tracking as **global min-cost flow over a graph**, replacing greedy frame-by-frame matching with a globally optimal association over the whole sequence. Relation (same family as your method): The clearest precedent for **graph-based temporal association**; your work adapts this graph philosophy to crown polygons with explicit split/merge edges and consensus-crown extraction, rather than a single min-cost-flow person-tracking objective.

bibtex

```
@inproceedings{zhang2008global,
  author    = {Zhang, Li and Li, Yuan and Nevatia, Ramakant},
  title     = {Global data association for multi-object tracking using network f
  booktitle = {2008 IEEE Conference on Computer Vision and Pattern Recognition (
  year      = {2008},
  pages     = {1--8},
  doi       = {10.1109/CVPR.2008.4587584}
}
```

THEME 2.5 — RGB-based phenology metrics

1. Sonnentag et al. (2012) — *Agricultural and Forest Meteorology* (digital repeat photography). **[Seminal — your GCC foundation]** Summary: Establishes GCC = G/(R+G+B) and ExG for canopy phenology and shows that, "due to its greater effectiveness in suppressing changes in scene illumination, especially in combination with per90, we advocate the use of gcc." Relation (same approach): Methodological foundation for your **GCC/RCC features** and for your explicit handling of **shadow/illumination** confounds via percentile/robust statistics. Scholars at the University ... Scholars at the University ...

bibtex

```
@article{sonnentag2012digital,
  author  = {Sonnentag, Oliver and Hufkens, Koen and Teshera-Sterne, Cory and Yo
  title   = {Digital repeat photography for phenological research in forest ecos
  journal = {Agricultural and Forest Meteorology},
  year    = {2012},
  volume  = {152},
  pages   = {159--177},
  doi     = {10.1016/j.agrformet.2011.09.009}
}
```

2. Richardson et al. (2018) — *Scientific Data* (PhenoCam Dataset). **[Seminal network methodology]** Summary: Presents the PhenoCam network methodology and a curated dataset from 393 cameras totaling nearly 1,783 site-years (e.g., 643 site-years for deciduous broadleaf forest), with standardized GCC processing and transition-date extraction. Relation (paradigm you scale down): The near-surface phenology paradigm you push to **individual-crown ROIs**; you differ by deriving per-crown ROIs from drone orthomosaics rather than fixed oblique tower cameras, avoiding low-LAI saturation from oblique views.

bibtex

```
@article{richardson2018tracking,
  author  = {Richardson, Andrew D. and Hufkens, Koen and Milliman, Thomas and Au
  title   = {Tracking vegetation phenology across diverse North American biomes
  journal = {Scientific Data},
  year    = {2018},
  volume  = {5},
  pages   = {180028},
  doi     = {10.1038/sdata.2018.28}
}
```

3. Richardson (2019) — *New Phytologist* (phenocam review). Summary: Reviews the "phenocam" approach and its applications, explicitly including close-up monitoring of individual organisms and canopy ROIs. Relation (justifies ROI-level tracking): Provides the conceptual license for **crown/ROI-level greenness and senescence tracking** at the individual-tree scale. **[verify exact pages]**

bibtex

```
@article{richardson2019tracking,
  author  = {Richardson, Andrew D.},
  title   = {Tracking seasonal rhythms of plants in diverse ecosystems with digi
  journal = {New Phytologist},
  year    = {2019},
  volume  = {222},
  number  = {4},
  pages   = {1742--1750},
  doi     = {10.1111/nph.15591},
  note    = {Verify pages against publisher}
}
```

4. Woebbecke, Meyer, Von Bargen & Mortensen (1995) — *Transactions of the ASAE* (ExG). **[Seminal index]** Summary: Introduces excess green (ExG = 2g − r − b) and related chromatic-coordinate indices that give a near-binary plant-vs-background image under varying soil, residue, and lighting. Relation (origin of your vegetation features): Seminal source of the **ExG / vegetation-fraction** features in your feature set and the rationale for chromatic-coordinate normalization. ScienceDirect

bibtex

```
@article{woebbecke1995color,
  author  = {Woebbecke, David M. and Meyer, George E. and Von Bargen, K. and Mor
  title   = {Color indices for weed identification under various soil, residue,
  journal = {Transactions of the ASAE},
  year    = {1995},
  volume  = {38},
  number  = {1},
  pages   = {259--269},
  doi     = {10.13031/2013.27838}
}
```

5. Liu et al. (2020) — *Agricultural and Forest Meteorology* (RCC). Summary: Shows the **red chromatic coordinate (RCC)** characterizes the phenology of canopy photosynthesis and complements GCC, performing well for end-of-season/senescence at deciduous sites. Relation (supports RCC use): Directly justifies your inclusion of **RCC alongside GCC** for detecting leaf-on/leaf-off and senescence in deciduous trees. **[verify volume/pages]**
ScienceDirect ScienceDirect

bibtex

```
@article{liu2020using,
  author  = {Liu, Ying and Wu, Chaoyang and Sonnentag, Oliver and Desai, Ankur R
  title   = {Using the red chromatic coordinate to characterize the phenology of
  journal = {Agricultural and Forest Meteorology},
  year    = {2020},
  volume  = {285--286},
  pages   = {107910},
  doi     = {10.1016/j.agrformet.2020.107910},
  note    = {Verify volume/pages against publisher}
}
```

6. Keenan et al. (2014) — *Ecological Applications* (critical assessment). Summary: Critically assesses greenness-index reliability, including LAI saturation, spring "greenness peak" artifacts, and illumination effects on digital-repeat-photography phenology. Relation (your caveats): Anchors your **caveats on shadow/illumination confounds** and on the limits of greenness indices, motivating your additional texture/entropy/Laplacian features.
Harvard University ScienceDirect

bibtex

```
@article{keenan2014tracking,
  author  = {Keenan, Trevor F. and Darby, Bijan and Felts, Erika and Sonnentag,
  title   = {Tracking forest phenology and seasonal physiology using digital rep
  journal = {Ecological Applications},
  year    = {2014},
  volume  = {24},
  number  = {6},
  pages   = {1478--1489},
  doi     = {10.1890/13-0652.1}
}
```

7. Toomey et al. (2015) — *Ecological Applications* (greenness vs photosynthesis). Summary: Demonstrates that camera-derived greenness indices predict the timing and seasonal dynamics of canopy-scale photosynthesis across many sites. Relation (validates the proxy): Validates greenness indices as ecologically meaningful **phenology proxies** that your rule-based deciduous/evergreen and leaf-on/off classifier relies upon. **[verify volume/pages]**

ResearchGate

bibtex

```
@article{toomey2015greenness,
  author  = {Toomey, Michael and Friedl, Mark A. and Frolking, Steve and Hufkens
  title   = {Greenness indices from digital cameras predict the timing and seaso
  journal = {Ecological Applications},
  year    = {2015},
  volume  = {25},
  number  = {1},
  pages   = {99--115},
  doi     = {10.1890/14-0005.1},
  note    = {Verify volume/pages against publisher}
}
```

THEME 2.6 — Field labels, species/trait mapping, and crown-level ecological interpretation

1. Ramaswami, Sidhu & Quader (2021) — *Current Science* (SeasonWatch). **[Seminal Indian citizen science]** Summary: Describes the India-wide SeasonWatch citizen-science program and builds baseline flowering/fruiting/leaf-flush seasonality for common tropical tree species from weekly volunteer observations. Relation (your field/citizen-science context): The directly relevant **Indian phenology citizen-science** reference grounding your IIT Delhi sites and your paper-form field-label collection; your work complements its coarse,

observer-based phenophases with **fine-grained drone-derived** crown phenology.
ResearchGate GBIF

bibtex

```
@article{ramaswami2021using,
  author  = {Ramaswami, Geetha and Sidhu, Swati and Quader, Suhel},
  title   = {Using citizen science to build baseline data on tropical tree pheno
  journal = {Current Science},
  year    = {2021},
  volume  = {121},
  number  = {11},
  pages   = {1409--1416},
  doi     = {10.18520/cs/v121/i11/1409-1416}
}
```

2. Schwartz, Betancourt & Weltzin (2012) — *Frontiers in Ecology and the Environment* (USA-NPN). **[Seminal network]** Summary: Traces the development of the USA National Phenology Network and its Nature's Notebook observation program from earlier lilac-network phenology efforts. Relation (international counterpart): The USA-NPN counterpart to SeasonWatch; situates your **field-annotation workflow** within established structured phenology-monitoring networks.

bibtex

```
@article{schwartz2012caprios,
  author  = {Schwartz, Mark D. and Betancourt, Julio L. and Weltzin, Jake F.},
  title   = {From Caprio's lilacs to the USA National Phenology Network},
  journal = {Frontiers in Ecology and the Environment},
  year    = {2012},
  volume  = {10},
  number  = {6},
  pages   = {324--327},
  doi     = {10.1890/110281}
}
```

3. Wu et al. (2021) — *ISPRS Journal of Photogrammetry and Remote Sensing* (drone + PlanetScope). **[Best precedent for your companion goal]** Summary: Integrates near-daily 3 m PlanetScope satellite imagery with drone observations to monitor **tree-crown-scale** autumn leaf phenology in a temperate forest. Relation (your future/companion work, gap you address): The strongest precedent for **using fine drone-derived crown phenology to inform coarser satellite phenology** — exactly your deferred companion goal; your contribution is an automated, temporally stable crown pipeline that could supply such fine labels at scale. ScienceDirect

bibtex

```
@article{wu2021monitoring,
  author  = {Wu, Shengbiao and Wang, Jing and Yan, Zhengbing and Song, Guangqin
  title   = {Monitoring tree-crown scale autumn leaf phenology in a temperate fo
  journal = {ISPRS Journal of Photogrammetry and Remote Sensing},
  year    = {2021},
  volume  = {171},
  pages   = {36--48},
  doi     = {10.1016/j.isprsjprs.2020.10.017}
}
```

4. Zhang et al. (2020) — *Journal of Forestry Research* (UAV-RGB species classification). Summary: Per the paper, "ResNet-50 achieved an overall accuracy (OA) of 92.6% and a kappa coefficient of 0.91" for crown-level tree-species classification from UAV RGB imagery, outperforming AlexNet and VGG-16. Relation (supports species-label join): Demonstrates the **crown → species** labeling task you perform by joining field species labels to detected crowns; differs in that you use field-collected labels rather than fully learned classification. **[verify author first names against DOI]** Springer Springer

bibtex

```
@article{zhang2020tree,
  author  = {Zhang, Chen and Xia, Kaifeng and Feng, Hanbo and Yang, Yanhao and D
  title   = {Tree species classification using deep learning and RGB optical ima
  journal = {Journal of Forestry Research},
  year    = {2020},
  volume  = {32},
  number  = {5},
  pages   = {1879--1888},
  doi     = {10.1007/s11676-020-01245-0},
  note    = {Verify author given names and end page against publisher}
}
```

5. Onishi & Ise (2021) — *Scientific Reports* (explainable UAV tree mapping). Summary: Object-based CNN classification of automatically segmented UAV-RGB crowns into seven tree classes at >90% accuracy, with Grad-CAM interpretation showing the network keys on crown shape and leaf contrast. Relation (crown→label workflow): Illustrates an end-to-end **crown segmentation + labeling** workflow analogous to your detection-then-label step. **[verify volume/article number/DOI]** nih

bibtex

```
@article{onishi2021explainable,
  author  = {Onishi, Masanori and Ise, Takeshi},
  title   = {Explainable identification and mapping of trees using UAV RGB image
  journal = {Scientific Reports},
  year    = {2021},
  volume  = {11},
  pages   = {903},
  doi     = {10.1038/s41598-020-79653-9},
  note    = {Verify article number/DOI against publisher}
}
```

6. Zhao et al. (2022) — *Remote Sensing of Environment* (fine-scale satellite phenology vs ground). Summary: Evaluates fine-scale phenology derived from PlanetScope CubeSat imagery against ground observations across temperate forests in eastern North America. Relation (reinforces upscaling pathway): Reinforces the **fine-to-coarse validation pathway** central to your companion goal — drone/ground labels validating or training satellite land-surface phenology. Usanpn

bibtex

```
@article{zhao2022evaluating,
  author  = {Zhao, Yingyi and Lee, Calvin K. F. and Wang, Zhihui and Wang, Jing
  title   = {Evaluating fine-scale phenology from PlanetScope satellites with gr
  journal = {Remote Sensing of Environment},
  year    = {2022},
  volume  = {283},
  pages   = {113310},
  doi     = {10.1016/j.rse.2022.113310}
}
```

Recommendations

1. **Lead each theme with the named seminal work**, then contemporary studies (the ranking above already does this). In Theme 2.4 specifically, structure the prose as: SORT → DeepSORT → Kuhn/Munkres (establish the Hungarian one-to-one paradigm and its split/merge failure modes) → Zhang/Li/Nevatia (graph/network-flow association) → **your method** as the crown-adapted graph association with consensus/medoid crowns. This is where your novelty lands hardest.
2. **In Theme 2.1 and 2.6, make the temporal-stability and label-transfer gaps explicit.** State plainly that Klosterman & Richardson (2017), Berra (2019), and Wu (2021) draw or assume per-date ROIs without an automated cross-date crown-linking step — that one sentence positions your consensus-crown contribution.

3. **Resolve the flagged fields before submission.** Priority `[verify]` items: the GIScience & RS urban multi-sensor author list, the *Drones* review author list, the Berra 2019 volume/pages, the ITCD-review final venue, and the page ranges for Richardson 2019, Liu 2020, Toomey 2015, Feurer & Vinatier 2018, and Zhang 2020. Each takes one lookup on the publisher DOI page.
4. **Thresholds that would change the shortlist:** if a target venue caps references, drop the two interchangeable "review backbone" items first (the *Drones* platforms review in 2.1 and the ITCD review in 2.2) before any seminal or directly-methodological source. If a reviewer demands a stronger urban-phenology precedent, prioritize adding an urban-tree phenology study over a second generic UAV review.

Caveats

- **Page/DOI verification:** I confirmed authors, year, venue, and DOI for every seminal work and most contemporary items directly from publisher/Crossref-backed sources; items marked `[verify]` have at least one field (usually pages or full author list) that I could not cross-confirm to two independent sources within budget — treat those as provisional and verify, do not fabricate.
- **Two genuinely uncertain author lists** (GIScience & RS urban multi-sensor 2021; *Drones* UAV-RS review 2023) are deliberately left as `{[verify authors]}` rather than guessed — fill them from the DOI page.
- **One paper, two roles:** Wu et al. (2021) and the tree-species CNN paper sit at the Theme 2.2/2.6 boundary; I placed each once to avoid double-counting. If you cite the species paper in both detection and labeling discussions, use a single BibTeX key.
- **Klosterman year:** the widely cited drone-phenology paper appears in *Sensors* 2017 (vol. 17, art. 2852); some later papers loosely cite it as "Klosterman et al. 2018." Use 2017 with the DOI given to be correct.
- **Munkres DOI:** `10.1137/0105003` is the standard SIAM DOI for the 1957 paper; confirm if your style requires the journal's modern name (J. SIAM → SIAM J.).